

Anthropometric measurement of the Chinese elderly living in the Beijing area

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Abstract

Anthropometric data of the elderly have become an immediate need for ergonomic design of health care and living products even in a developing country like China. The first aim of this survey was to collect anthropometric data of the Chinese elderly (aged over 65) living in the Beijing area. 58 females (age range 65.0–80.7, mean 71.2, SD 4.1) and 50 males (age range 65.2–85.1, mean 71.5, SD 4.4) took part in the survey. A total of 47 anthropometric dimensions and three items of functional strength were measured. Mean values, standard deviations, coefficients of variation, and percentiles for each parameter were estimated. It was found that in most dimensions there were no significant differences between the age groups of 65–69 and 70–74 or between the age groups of 70–74 and 75+. Male and female elderly had no significant differences in the body dimensions around the hip area. Comparison between Chinese (Beijing) and Japanese elderly shows that Chinese (Beijing) elderly are larger in the dimensions of the body trunk, and Japanese elderly are larger in the dimensions of the head and extremities. The conclusions are based on a limited number of subjects in the Beijing area, and the in-depth reasons for the above findings remain a subject for further study.

Relevance to industry

The continuous growth of the number of aged people has created a big market of health care and living products for the elderly. Anthropometric data are essential to the ergonomic design of these products. However, available anthropometric data for aged people are quite limited. This study fills part of this gap by supplying anthropometric data of the Chinese elderly.

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1. Introduction

The number of elderly people and their percentage of the whole population are greater than ever. According to the data of the United Nations, in 2002, the worldwide total number of people aged 60 and older was 629 million (10% of the whole population), and by the year 2050 it will increase to 1.964 billion (21% of the whole population) (ECOSOC, 2002). In China, the ageing process of the population is similar. In 1982, the number of elderly people was 76 million;

in 2002, it amounted to 134 million. The percentage of elderly people in the whole population of China increased from 7.64% in 1982 to 10.0% in 2002. According to predictions, the current trend of the ageing of the Chinese population will continue. By the year 2010, the percentage of elderly people in the population will have grown to 12.5%, and in the year 2050 to 30.0% (CNCA, 2002).

With the non-reversing ageing trends, it is necessary to consider elderly people's physical limitations in the design of products for daily use; to do this, their anthropometric characteristics must be quantified. According to Jarosz (1999), the development of anthropometric research of elderly people can be divided into two phases. The first

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phase took place during the period of the 1960s, and most of the investigations were conducted in the United States and Western European countries. The first standard that defined the dimensions of the elderly for the needs of design was developed in Great Britain in 1969 (BS 4467, 1969). The second phase of anthropometric research of this population took place in the 1980s and 1990s. Interest in the anthropometric measurement of the elderly has grown in recent years.

The ageing process involves modifications in physiological and nutritional status (Perissinotto et al., 2002). Height, weight, body mass index (BMI) and muscle reserves decrease with increasing age (Corish and Kennedy, 2003). Muscle loss appears to be greater among men than women, and the reduction in muscle size has been associated with decreased handgrip strength (Corish and Kennedy, 2003; Pennathur and Dowling, 2003). Body composition changes occur differently in men and women and in the various phases of ageing, influencing anthropometry (Perissinotto et al., 2002; Suriah et al., 1998). Consequently, the anthropometric standard values derived from adult populations may not be applicable to the elderly. Non-pathological factors affecting the distribution of anthropometric characteristics, such as age, gender and geographic area should be taken into account (Kothiyal and Tetley, 2000). The WHO Expert Committee on Physical Status has stressed the need for local gender- and age-specific reference values for the elderly (de Onis and Habicht, 1996).

In China, anthropometric research of elderly people for design use has not been conducted. The available anthropometric data only includes subjects up to the age of 60 (GB10000-88, 1988). Therefore, the first aim of the present study was to provide anthropometric data of the elderly based on a sample drawn from the Beijing area; the second aim was to describe the gender and national (compared with Japanese data) differences of anthropometric characteristics in the elderly.

2. Methods

2.1. Subjects

The number of subjects was estimated according to the equation provided in Annex A of ISO 15535:2003 “General requirements for establishing anthropometric databases” for a 95% confidence interval for the 5th and 95th percentiles:

$$n \geq \left(3.006 \times \frac{CV}{\alpha} \right)^2, \quad (1)$$

where n is sample size, CV is the coefficient of variation, and α is the percentage of relative accuracy desired. In this survey, 10% relative accuracy is desired for the 5th and 95th percentiles and an empirical value of $CV = 25$ is used

to pre-determine the sample size. The result is about 56 for both male and female.

The present survey was carried out in the Beijing area. All subjects were in a state of good health, and those who were unable to stand unassisted were excluded from the study. Before the start of data collection, the subjects were told that the study was to develop an anthropometric database for the purpose of improving products or other ergonomic considerations for elderly people. The measurement procedure was also explained in detail to them. After they fully understood the purpose and the procedure of the study, they could decide whether or not to take part in the survey. Subjects agreeing to take part in the study were asked to sign a consent form and fill out a brief medical history form.

Subjects aged over 65 were recruited from the relevant group of retired people living in Beijing mainly in August 2004, considering the climate was comfortable for the aged to be lightly dressed. After removing data of a few subjects because their decimal ages were a little lower than 65, data of 58 women and 50 men were processed in later analysis. The birthplaces of the subjects were diverse, covering 17 provinces/municipalities. In the last century, the population structure changed greatly, especially after new China was established. The age range of the females was 65.0–80.7 with a mean of 71.2 and SD of 4.1, and the age range of the males was 65.2–85.1 with a mean of 71.5 and SD of 4.4. The age distribution was not balanced. The number of subjects in the age groups of 65–69, 70–74 and 75–79 were 45, 36 and 24, respectively. In the over 80 age group, there were three subjects altogether. Therefore, in the following discussion, the 75–79 and 80+ age groups are combined into a new group designated as 75+.

2.2. Dimension measurements

Dimensions were measured by trained experimenters using equipment including a weighing scale, two Martin-type anthropometers (Kroemer et al., 1986; Shao, 1985) for standing and sitting postures, respectively, a sliding caliper, a spreading caliper, a foot measurer and a plastic tape. The weighing scale was used to measure the body weight. During the measurement, subjects were required to wear only light clothing and no shoes. The Martin-type anthropometer for the standing posture is a widely used anthropometric device for measuring heights, breadths and depths between points on the body and standard reference surfaces in a standing posture. The Martin-type anthropometer for the sitting posture is similar to the one for the standing posture, but has a flat wooden seat with a high backrest. The seat and the backrest are aligned at a right angle to each other. The seat can be adjusted to different heights and acts as a reference point for the measurements in the sitting position. The sliding caliper and spreading caliper were used to measure small breadths and depths of body segments, such as head and hands. The foot measurer

was used to measure the dimensions of the foot, and the plastic tape was used to measure body circumferences.

An adequate description of the human body may require over 300 dimensions (Pheasant, 1986; Roebuck et al., 1975), and the scope of this study was limited to measurement of body dimensions that were considered important for facility design for the elderly. Forty seven body dimensions (24 in the standing posture and 23 in the sitting posture) were selected, as shown in the left column of Table 1. All dimensions were as defined in the Chinese

National Standard GB/T 5703 1999 (equivalent to ISO 7250:1996): basic human body measurements for technological design.

2.3. Functional strength measurements

A person's capacity to perform mechanical work is determined by his/her ability to exert muscular strength (Mital and Kumar, 1998a). It has been of interest in physiological and biomechanical research for many

Table 1
Descriptive statistics of dimension measurements (mm)

Dimensions	Male				Female			
	N	M	SD	CV (%)	N	M	SD	CV (%)
Weight (kg)	50	68	10.6	15.7	57	60	9.7	16.2
Stature	50	1655	54.3	3.3	55	1526	69.3	4.5
Eye height	50	1545	55.5	3.6	54	1418	65.5	4.6
Shoulder height	50	1376	54.4	4.0	55	1263	62.7	5.0
Middle fingertip height	50	627	44.1	7.0	54	576	44.1	7.6
Waist height	50	998	38.9	3.9	55	9329	63.7	6.8
Crystal iliac height	50	944	51.1	5.4	55	8975	51.9	5.8
Elbow height	50	1023	49.1	4.8	54	942	55.6	5.9
Functional hand height	50	726	37.9	5.2	55	665	56.6	8.5
Tibial height	49	468	30.7	6.6	55	434	27.1	6.2
Crotch height	50	720	45.6	6.3	55	681	65.8	9.7
Shoulder breadth	50	330	22.7	6.9	56	286	25.3	8.8
Maximum shoulder breadth	49	377	23.2	6.1	56	328	24.6	7.5
Hip breadth	50	344	19.9	5.8	55	345	21.0	6.1
Waist depth	50	258	39.3	15.3	54	260	29.2	11.2
Waist breadth	50	309	32.2	10.4	55	306	34.4	11.3
Crista iliaca breadth	50	310	27.5	8.9	55	321	27.8	8.7
Waist point circumference	50	887	109.6	12.4	54	881	95.4	10.8
Waist circumference	50	910	106.3	11.7	55	941	105.4	11.2
Hip circumference	50	963	61.4	6.4	54	973	76.7	7.9
Thigh circumference	50	505	57.8	11.4	55	520	46.6	9.0
Crotch length	49	766	56.9	7.4	55	726	72.5	10.0
Ankle height	48	66	7.8	11.9	56	59	9.6	16.4
Functional thumb-tip reach	49	762	48.4	6.4	54	721	39.8	5.5
Sitting height	50	879	29.4	3.3	58	807	44.3	5.5
Cervical height, sitting	50	644	33.3	5.2	57	594	54.0	9.1
Eye height, sitting	50	770	32.2	4.2	58	698	44.0	6.3
Shoulder height, sitting	48	604	30.4	5.0	58	552	35.3	6.4
Thigh clearance height, sitting	50	125	13.0	10.4	57	123	13.9	11.3
Elbow height, sitting	49	254	30.0	11.8	56	226	31.0	13.7
Forearm-fingertip length	50	455	27.7	6.1	56	431	24.9	5.8
Abdominal depth, sitting	50	265	42.7	16.1	56	291	33.9	11.7
Buttock-knee length, sitting	50	550	33.4	6.1	55	538	27.2	5.0
Sitting depth	50	444	32.8	7.4	57	448	36.6	8.2
Hip breadth, sitting	50	362	35.8	9.9	58	368	25.3	6.9
Knee height, sitting	49	484	26.9	5.6	57	457	31.3	6.8
Popliteal height	48	403	27.8	6.9	57	372	33.2	8.9
Maximum head breadth	49	156	7.1	4.6	57	150	7.4	4.9
Maximum head length	49	187	7.9	4.2	57	179	8.9	5.0
Interpupillary distance	50	61	4.0	6.5	57	60	5.1	8.5
Total head height	50	227	14.4	6.3	57	216	13.3	6.2
Hand breadth at metacarpal	50	84	4.5	5.4	58	78	4.2	5.3
Maximum hand breadth	50	102	6.3	6.2	58	95	5.5	5.8
Hand length	50	179	8.1	4.5	58	168	8.1	4.8
Finger length	48	69	4.7	6.8	58	66	4.1	6.3
Foot length	50	242	10.9	4.5	58	224	12.2	5.5
Foot breadth	50	91	8.1	8.9	56	86	4.8	5.6

years. In-depth discussions on human strength and its measurement have been presented by Chaffin and Andersson (1991), Mital and Kumar (1998a,b) and Kroemer (1999). Reduced muscle strength significantly influences living activities of the elderly. Unfortunately, there are few strength data for the elderly all around the world. In this survey, handgrip strength, back strength (or torso lifting strength) and leg strength (or leg lifting strength, composite strength) were selected as important strength items reflecting functional strength capability of the elderly.

For safety considerations, we measured maximum voluntary strength parameters of the elderly without any leg/back problems. Moreover, the subjects were not asked for an exertion duration of 4–6 s as suggested in the standardized procedures described in Chaffin and Andersson (1991) and Kroemer (1999) for general population measurement; instead, they were asked to exert to their voluntary peaks at a self-controlled slow speed no matter how long they could maintain at the peaks. Maximum voluntary handgrip strength, back strength and leg strength were measured using a Smedley's handgrip

dynamometer (CWL-I, Fig. 1(a), range: 0–100 kg, graduation: 1 kg) and a digital back/leg dynamometer (BCS-400, Fig. 1(b), range: 0–400 kg, graduation: 1 kg). The handgrip dynamometer has two parallel bars, and the position of the inner bar is adjustable so that the distance between the two bars can be set according to the hand size of the subject. Before measurement, the subject adjusted the bar distance so that the middle segments of his/her index, middle, ring and little fingers could grip the inner bar most effectively (subjective judgment). Then the pointer was set to the zero position. After the subject had gripped to his/her maximum voluntary handgrip strength, the pointer stayed at its final position, and the experimenter could then read and record the result.

The digital back/leg dynamometer had an adjustable handgrip chain and an LCD screen for quick and reliable reading. Before measurement, the experimenter explained and demonstrated the correct exercise method to the subject. The chain was adjusted so that the gripping bar met the knee height of the subject at the measured posture. For leg strength measurement, the subject stood on the platform with his/her feet a suitable distance apart (about shoulder width), grasped the gripping bar at both ends, flexed the knees to a favorable angle (about 135°), kept the back straight, the hips over the ankle joints, the chest forward and the head erect, and then slowly extended the knees smoothly (at the same time the hips rose vertically) to the maximum voluntary force.

For back strength measurement, the procedure was similar, but the legs (instead of the back) were kept straight, the body was bent forward around the hip joint, and the subject extended his/her trunk around the hip joint by pulling the bar upwards to maximum voluntary force (at the same time, the trunk reached its final angle), then the force value was recorded.

Each strength item was measured three times for each subject, and the highest of three readings was used as the result for each item.

2.4. Statistical analysis

The Statistical Package for the Social Sciences (SPSS) for Windows version 12.0 was used in the following statistical analysis. Extreme outliers, results that are unreasonable and probably come from mistakes in measurement or recording, were identified and eliminated carefully. Normality was examined by 1-sample K-S method, and then descriptive statistics, including arithmetic means (M), standard deviations (SD), and percentiles (5th, 50th, and 95th) of the above measurements were calculated for both the male and female population. Variation was also expressed by coefficient of variation (CV) values. Missing data were not replaced by estimated values. In comparison between population groups, an independent sample *T*-test ($\alpha = 95\%$) was adopted.

The relative accuracy α and standard error S_P for the 5th and 95th percentiles were estimated by the following

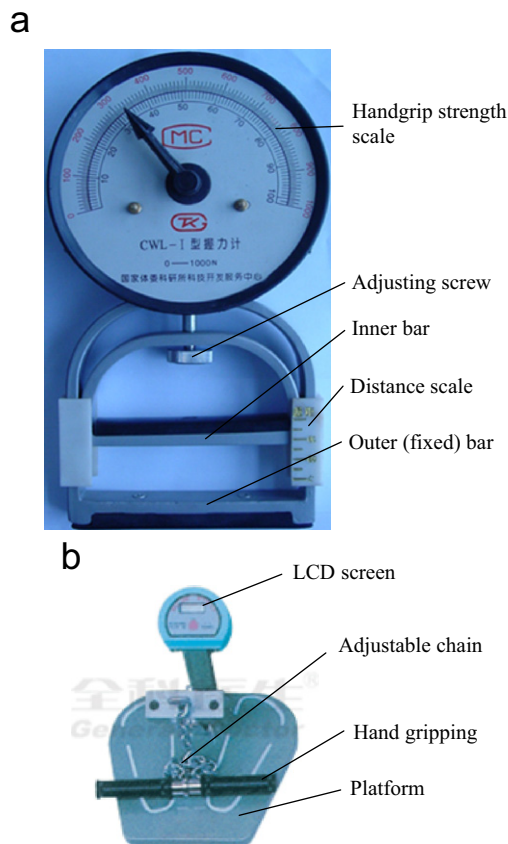


Fig. 1. Apparatus for strength measurement: (a) handgrip dynamometer (CWL-I); (b) digital back/leg dynamometer (BCS-400).

equations adapted from ISO 15535: 2003:

$$S_P = \frac{SD}{\sqrt{n}} \times 1.534, \quad (2)$$

$$\alpha = \frac{CV}{\sqrt{n}} \times 3.006. \quad (3)$$

3. Results

3.1. Anthropometric data of Chinese elderly living in Beijing

Table 1 presents the descriptive statistics of the anthropometric dimensions for the males and females, including the number of subjects, mean, SD, and CV. Table 2 presents percentile values for the anthropometric dimensions of both male and female subjects.

The sample sizes for various measurements were different. For some measurements, some subjects could not be measured properly because of their physical weakness on exercising the required posture. Even to stature measurement, some elderly were found unable to stretch their body as required. The measurements under such situations were skipped. Also, if we found that the health of a subject was not good during measurement, we stopped the measurement of this subject. Some data were found wrongly recorded and thus were deleted.

Table 1 shows that, for dimension measurements, most CVs were far lower than we had assumed (25%) and no CVs were greater than the assumption. The relative accuracy and standard error for the 5th and 95th percentiles were estimated according to Eqs. (2) and (3). For female subjects, the relative accuracy ranged from 1.8% (stature) to 8.3% (popliteal height) with a mean of 3.3%, and the standard error for the 5th and 95th dimension percentiles ranged from 0.8 mm (finger II length) to 21.8 mm (waist circumference) with a mean of 7.9 mm. For male subjects, the relative accuracy ranged from 1.4% (stature) to 6.9% (abdominal depth, sitting) with a mean of 3.2%, and the standard error for the 5th and 95th dimension percentiles ranged from 0.9 mm (interpupillary distance) to 23.8 mm (waist point circumference) with a mean of 7.5 mm.

Great gaps existed in functional strengths between the male and female subjects, as shown in Table 3. Males had almost twice the strength of females. All three types of strengths had high CV values, which shows that the differences among individuals were great. The CV of leg strength reached 38% (the corresponding relative accuracy for the 5th and 95th percentiles was 15%). Unlike body dimensions, strength can be influenced by many factors, such as physical activity. This may partially explain why the CV values of strength measurements were so high. Table 4 presents the percentile values for the functional strength of both male and female subjects.

It should be pointed out that the measurements were not conducted at the same time for all subjects (because it was

not practical to do this), and thus diurnal variations exist in the data.

4. Discussion

4.1. Differences between age groups

Measuring height reliably in the elderly is one of the most problematic areas of anthropometry (Corish and Kennedy, 2003). In old age, there is a decline in sitting and standing height due to vertebral compression, change in the height and shape of the vertebral discs, loss of muscle tone and postural changes (WHO, 1995). Although various methods to estimate height in the elderly have been developed, their validity remains uncertain. Standing height (stature) is still the most widely used statistic for height measurement in the elderly (Van Staveren et al., 1995; Rea et al., 1997; Finch et al., 1998) and was measured by a standard method in the present study. Table 5 shows that the subjects in the older groups had lower mean statures. The difference between the 65–69 group and the 70–74 group was 33 mm for females and 47 mm for males, and the difference between the 70–74 group and the 75+ group was 8 mm for females and 5 mm for males. However, *T*-test results indicated that none of the differences were significant. There were significant differences between the 65–69 group and the 70–74 group for the following dimensions: elbow height, functional thumb-tip reach, forearm-plus-hand length, buttock-knee length (sitting), maximum hand breadth, hand length, and finger II length for males; and elbow height (sitting) and functional thumb-tip reach for females. There were significant differences between the 70–74 group and the 75+ group only in maximum shoulder breadth for females, but no significant differences were found for males. Most of the dimensions with significant differences were not related to vertebral compression. The exact reasons for the significant differences remain unknown; we could not identify them in this study.

4.2. Gender differences

The significance of the differences between male and female subjects was examined by *T*-test.

No significant differences were found between male and female subjects in the dimensions of hip breadth, waist depth, waist breadth, waist point circumference, waist circumference, hip circumference, thigh circumference, or interpupillary distance in the standing posture. And there were no significant differences in the dimensions of sitting thigh clearance height, sitting depth, and sitting hip breadth in the sitting posture. With the exception of interpupillary distance, all of these dimensions in standing and sitting postures were around the hip area. The results showed that there was very little difference in the hip area between elderly males and females, which may imply that it is not necessary to consider gender difference in the design

Table 2
Percentile values of dimension measurements (mm)

Dimensions	Male					Female				
	1st	5th	50th	95th	99th	1st	5th	50th	95th	99th
Weight (kg)	48	51	67	88	91	33	44	59	77	78
Stature	1540	1549	1657	1770	1782	1320	1403	1522	1628	1645
Eye height	1408	1441	1548	1657	1679	1237	1294	1415	1522	1532
Shoulder height	1254	1279	1376	1482	1508	1109	1146	1264	1351	1399
Middle fingertip height	538	550	627	723	730	462	483	583	640	657
Waist height	899	919	997	1066	1102	641	832	939	1022	1032
Crystal iliac height	818	830	955	1028	1046	765	810	894	986	999
Elbow height	929	942	1024	1129	1150	801	833	945	1029	1041
Functional hand height	637	652	731	790	829	527	537	679	757	762
Tibial height	388	412	467	510	591	347	390	436	473	484
Crotch height	548	650	727	795	815	364	576	685	759	856
Shoulder breadth	264	285	330	365	381	206	241	290	325	338
Maximum shoulder breadth	320	325	379	412	424	270	283	333	366	376
Hip breadth	291	295	346	373	382	300	307	348	381	397
Waist depth	196	203	253	340	358	200	211	261	308	341
Waist breadth	244	256	310	374	396	214	239	307	360	377
Crista iliaca breadth	261	267	312	357	386	266	283	320	376	383
Waist point circumference	610	706	892	1073	1118	646	716	880	1039	1056
Waist circumference	690	721	926	1074	1168	634	721	955	1100	1133
Hip circumference	800	853	963	1067	1072	612	843	980	1081	1085
Thigh circumference	245	245	521	582	585	384	445	520	587	625
Crotch length	645	675	765	870	930	580	622	720	886	970
Ankle height	53	54	66	84	90	33	37	60	73	75
Functional thumb-tip reach	559	672	770	824	842	645	660	722	795	816
Sitting height	807	826	881	929	935	669	716	812	873	916
Cervical height, sitting	573	587	644	701	755	456	531	593	670	886
Eye height, sitting	675	718	770	824	833	566	601	702	758	795
Shoulder height, sitting	537	557	605	660	695	460	477	557	601	666
Thigh clearance height, sitting	95	101	125	145	150	85	102	123	145	152
Elbow height, sitting	178	197	254	308	314	148	174	224	278	296
Forearm-fingertip length	387	397	459	490	499	331	390	432	464	493
Abdominal depth, sitting	178	186	262	331	355	210	224	292	349	381
Buttock-knee length, sitting	483	491	551	604	626	481	482	533	589	597
Sitting depth	364	388	446	495	501	307	394	447	514	576
Hip breadth, sitting	260	284	362	429	430	292	327	375	404	410
Knee height, sitting	430	432	483	534	540	370	389	458	504	511
Popliteal height	320	359	406	450	467	266	293	376	415	419
Maximum head breadth	142	144	157	169	174	132	138	150	161	178
Maximum head length	171	173	187	204	206	150	157	180	192	196
Interpupillary distance	53	55	61	68	71	46	52	61	69	71
Total head height	190	202	227	249	258	185	192	216	242	255
Hand breadth at metacarpal	73	77	83	92	93	67	73	78	85	92
Maximum hand breadth	89	91	102	113	117	77	87	94	102	108
Hand length	163	164	179	193	197	143	159	166	183	187
Finger length	60	61	70	79	83	57	60	66	75	75
Foot length	201	226	241	260	262	184	200	224	243	252
Foot breadth	63	67	93	101	103	75	78	87	94	99

Table 3
Descriptive statistics of strength measurements (kg)

Strength measurements	Male				Female			
	N	M	SD	CV (%)	N	M	SD	CV (%)
Grip	49	39	11	28.2	55	21	5.2	24.7
Back strength	49	70	19	27.1	55	35	14	40.0
Leg strength	49	84	33	39.3	53	39	15	38.5

Table 4
Percentile values of the strength measurements (kg)

Strength measurements	Male			Female		
	5th	50th	95th	5th	50th	95th
Grip	23	38	53	13	21	30
Back strength	38	71	103	15	34	59
Leg strength	29	86	141	16	40	66

of some health care products such as pull-on incontinence underwear.

4.3. Differences between Chinese (Beijing) and Japanese elderly

Chinese and Japanese people both belong to the Mongolian races and are highly associated historically. It will be interesting to find out whether there are significant differences in body dimensions of the elderly in these two groups under different social and economic situations. It is also meaningful for business, since many companies are considering transferring their products for the Japanese market to the Chinese market, especially to large Chinese cities like Beijing.

The anthropometric data of the Chinese (Beijing) and Japanese elderly for 17 anthropometric dimensions are presented in Table 6. The Japanese data were collected during 1997 and 1998 by the National Institute of Technology and Evaluation (NITE) and the National Institute of Advanced Industrial Science and Technology (AIST), Japan (a CD-ROM with the data and descriptions is available for purchase).

Table 5
Comparison of stature among age groups (mm)

Age group	Male			Female		
	N	M	SD	N	M	SD
65–69	19	1674	54	24	1544	60
70–74	18	1646	56	17	1521	60
75+	13	1640	48	14	1499	87

Table 6
Comparison between Chinese and Japanese elderly in 17 anthropometric dimensions (mm)

	Male						Female					
	Chinese			Japanese			Chinese			Japanese		
	N	M	SD	N	M	SD	N	M	SD	N	M	SD
Weight (kg)	50	68	11	41	61	8	57	60	10	30	54	8
Stature	50	1655	54	41	1609	58	55	1525	69	30	1475	47
Waist height	50	998	39	41	959	41	55	932	64	30	891	34
Crystal iliac height	50	944	51	41	860	35	55	897	52	30	784	32
Maximum shoulder breadth	50	330	23	41	377	16	56	286	25	30	347	14
Waist breadth	50	309	32	41	293	21	55	305	34	30	283	30
Waist circumference	50	910	106	40	854	78	55	941	105	30	812	92
Hip circumference	50	964	61	41	893	51	54	973	77	30	921	65
Thigh circumference	50	505	58	41	491	38	55	520	47	29	517	40
Ankle height	49	66	8	41	67	4	57	59	10	30	58	4
Maximum head breadth	49	156	7	40	158	6	57	150	7	30	152	5
Maximum head length	50	187	8	41	192	6	57	179	9	30	182	5
Total head height	50	227	14	41	240	8	58	216	13	30	231	7
Hand breadth at metacarpal	50	84	5	41	86	5	58	78	4	30	79	4
Hand length	50	179	8	41	177	8	58	168	8	30	178	7
Foot length	48	242	11	41	243.4	11	56	224	12	30	228	9
Foot breadth	50	91	8	41	104	7	56	86	5	30	97	5

The *T*-test was used again to analyze the significances of differences in body dimensions between the Chinese (Beijing) and Japanese elderly. It was found that Chinese (Beijing) aged females had greater values in weight, stature, crystal iliac height, waist circumference, hip circumference, waist breadth and waist height, whereas Japanese aged females had greater values in maximum shoulder breadth, head height, hand length, foot breadth and foot length. There were no significant differences between Chinese and Japanese females in hand breadth at metacarpal, maximum head length, maximum head breadth, ankle height and thigh circumference. Chinese aged males had higher measurements in weight, stature, waist circumference, waist breadth, waist height, and hip circumference, whereas Japanese aged males had higher values in maximum shoulder breadth, total head height, maximum head length, foot breadth and hand breadth at the metacarpal. There were no significant differences between Chinese and Japanese males in ankle height, thigh circumference, maximum head breadth, foot length and hand length. Overall, Chinese aged males and females have larger body trunks, and Japanese aged males and females have larger heads and extremities.

4.4. Comparison among different literatures

Table 7 lists several comparisons of the results of previous anthropometric studies of the elderly, including the nationality, the age ranges and the numbers of subjects and the number of dimensions measured. The comparisons of weight, stature and sitting height among these studies are also listed in Table 7, which shows some remarkable anthropometric differences between nations/

Table 7
Comparison among different literatures

No.	Author	Nation	Age range of subjects	Number of subjects	Number of dimensions measured	Mean weight (kg)		Mean stature (mm)		Mean sitting height (mm)	
						Male	Female	Male	Female	Male	Female
1	Kothiyal and Tetley (2000)	Australia	65–92 (male) 65–92 (female)	33 (male) 138 (female)	22	72	61	1658	1521	843	784
2	Perissinotto et al. (2002)	Italy	65–84 (male and female)	5462 (male and female)	4	72.6	63.8	1717	1522	N/A	N/A
3	Suriah et al. (1998)	Malaysia	60–89 (male and female)	140 (male) 204 (female)	4	58.66	51.69	1582.4	1474.4	N/A	N/A
4	Jarosz (1999)	Poland	60+ (female)	106 (female)	33	N/A	65.4	N/A	1524.8	N/A	798.9
5	Kuczmarski et al. (2000)	US	50+ (male and female)	7561 (male and female)	4	86	70.9	N/A	N/A	N/A	N/A
6	Pennathur and Dowling (2003)	US	60–85 (male and female)	40 (male) 106 (female)	15	N/A	N/A	1664.3	1525.7	800.9	756.6
7	Corish and Kennedy (2003)	Ireland	65+ (male and female)	276 (male) 598 (female)	5	77.81	65.31	1705.3	1565.3	N/A	N/A
8	NITE and AIST	Japan	65+	41 (male) 30 (female)	49	61	54	1609	1475	N/A	N/A
9	Hu et al. (this paper)	China	65–85 (male) 65–80 (female)	53 (male) 60 (female)	47	68	60	1655	1525	879	806

surveys. Note that there is a slight difference in the subject age ranges.

5. Conclusions

This study was conducted to provide anthropometric information of Chinese elderly subjects aged 65 years and above in the Beijing area, which could be used for the ergonomic design of working and living environment/products. A total of 47 anthropometric dimensions and three functional strengths are listed in the forms of mean, standard deviation, coefficient of variation and percentile values. The differences among age groups, between male and female groups, and between Chinese and Japanese elderly are discussed. The results show that the difference between male and female elderly in the hip area is not significant. Chinese male and female elderly have larger body trunks than Japanese, and Japanese male and female elderly are larger in the head and extremities. In this study, the sample size of the elderly was limited due to time and financial limits. In addition, the subjects were chosen based on convenience and availability; the age distribution could not be controlled, and elderly Chinese outside the Beijing area were not included in the study. Nevertheless, this survey provides the first anthropometric database of Chinese elderly aged over 65.

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